

Confidence Intervals and Confidence Regions

From scalar pivots to simultaneous ellipsoidal uncertainty

$$\hat{\beta} \pm t SE(\hat{\beta})$$



joint region

Learning objectives

Confidence statements are probability statements about procedures, not probabilities about fixed parameters.

ROADMAP

- Derive exact confidence intervals for estimable linear functions of β under the normal linear model.
- Explain the test–confidence-set duality and use it to construct confidence regions.
- Derive ellipsoidal confidence regions for β and for $C\beta$.
- Distinguish pointwise, joint, and simultaneous coverage.
- Construct intervals for a mean response and a future observation, and explain why prediction is wider.
- Recognize how rank deficiency changes the target: identifiable estimable functions remain inferentially meaningful.

From hypothesis tests to confidence sets

The same pivots and quadratic forms generate both procedures.

BIG PICTURE

Testing

Fix a null value and ask whether the observed statistic is unusually far from that value.

Reject values that are incompatible with the data.

Confidence set

Collect all parameter values that would not be rejected by the corresponding level- α test.

The result is a set of plausible values.

$$\{\theta_0 : \text{do not reject } H_0 : \theta = \theta_0\}$$

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Scalar confidence intervals

Start with one estimable linear function, then build toward joint regions.

Normal linear model setup

SETUP

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I), \quad \text{rank}(X) = p$$

Then $\hat{\beta} = (X^T X)^{-1} X^T Y \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$

This multivariate normal result is the engine behind both scalar intervals and joint regions.

Key question: how do we turn this sampling distribution into a set that covers the unknown parameter with probability $1 - \alpha$?

A linear function of β

TARGET

For any fixed $c \in \mathbb{R}^p$, consider $\theta = c^T \beta, \quad \hat{\theta} = c^T \hat{\beta}$

$$c^T \hat{\beta} \sim N(c^T \beta, \sigma^2 c^T (X^T X)^{-1} c)$$

Examples

- a single coefficient β_j
- a contrast $\beta_1 - \beta_2$
- a mean response $x_0^T \beta$
- any estimable linear combination

Standard error structure

The design enters only through

$$c^T (X^T X)^{-1} c.$$

Larger values mean less information in direction c .

Known σ : the normal pivot

KNOWN VARIANCE

$$Z = \frac{c^T \hat{\beta} - c^T \beta}{\sigma \sqrt{c^T (X^T X)^{-1} c}} \sim N(0, 1)$$

Therefore, with probability $1 - \alpha$,

$$c^T \beta \in c^T \hat{\beta} \pm z_{1-\alpha/2} \sigma \sqrt{c^T (X^T X)^{-1} c}$$

Exact finite-sample interval under the normal model.

Unknown σ : replace σ by the residual standard error

T PIVOT

$$s^2 = MSE = \frac{SSE}{n-p} \quad \text{From Cochran's theorem: } \frac{(n-p)s^2}{\sigma^2} \sim \chi_{n-p}^2, \quad \hat{\beta} \perp s^2$$

$$T = \frac{c^T \hat{\beta} - c^T \beta}{s \sqrt{c^T (X^T X)^{-1} c}} \sim t_{n-p}$$

$$c^T \hat{\beta} \pm t_{n-p, 1-\alpha/2} s \sqrt{c^T (X^T X)^{-1} c}$$

This is the familiar regression confidence interval, now derived from the linear-model geometry.

Why the t pivot appears

Normal numerator + independent chi-squared variance estimate = Student t.

DERIVATION

1 Normal contrast



2 Residual quadratic form



3 Independence



4 Student t

$$\frac{c^T \hat{\beta} - c^T \beta}{\sigma \sqrt{v_c}} \sim N(0, 1) \quad \frac{(n-p)s^2}{\sigma^2} \sim \chi_{n-p}^2 \quad \hat{\beta} \perp s^2 \quad \frac{c^T \hat{\beta} - c^T \beta}{s \sqrt{v_c}} \sim t_{n-p}$$

where $v_c = c^T (X^T X)^{-1} c$.

Worked example: interval for a contrast

EXAMPLE

Suppose the fitted model gives

$$\begin{aligned}c^T \hat{\beta} &= 2.40 \\s &= 1.50 \\c^T (X^T X)^{-1} c &= 0.16 \\n - p &= 24\end{aligned}$$

For a 95% interval, $t_{0.975, 24} \approx 2.064$.

$$SE(c^T \hat{\beta}) = 1.50 \sqrt{0.16} = 0.60$$

$$2.40 \pm 2.064(0.60) = 2.40 \pm 1.238$$

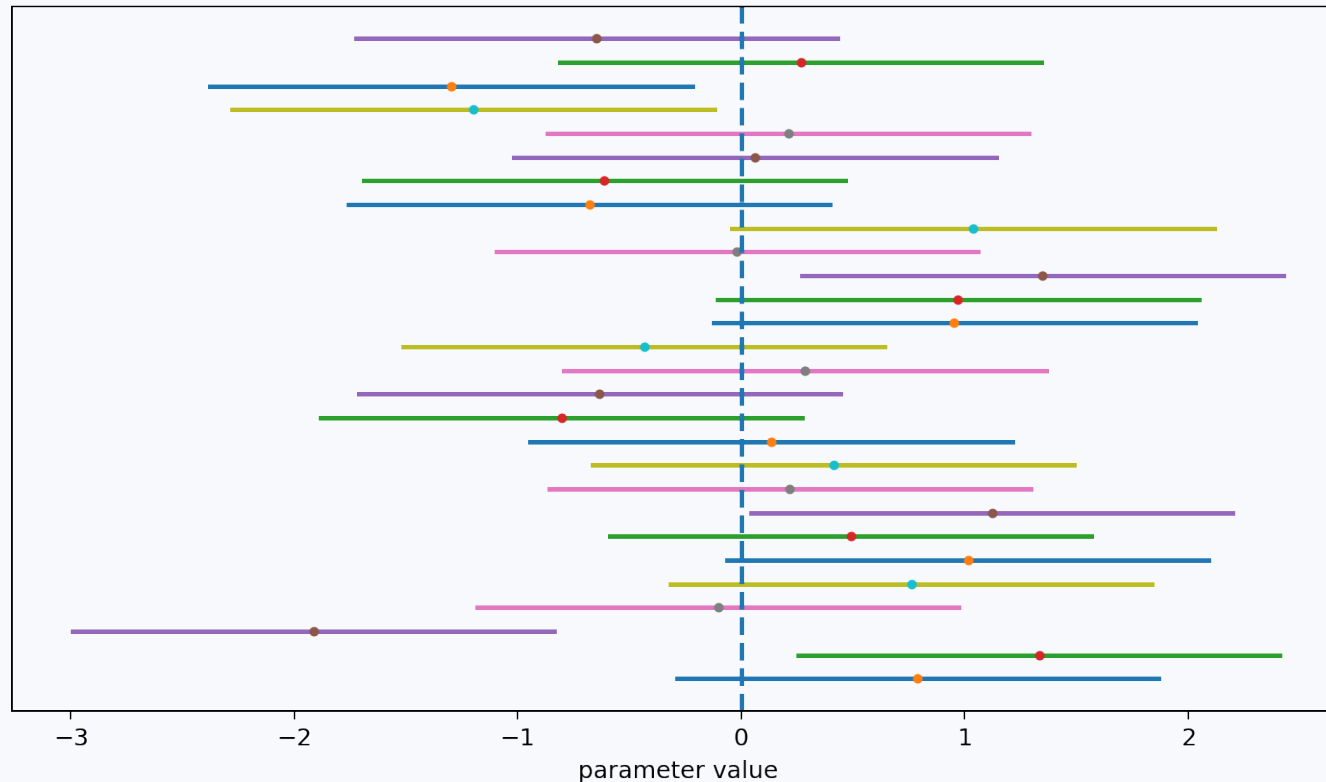
95% CI: (1.16, 3.64)

Interpretation: the procedure covers the fixed contrast $c^T \beta$ in 95% of repeated samples under the model.

What does “95% confidence” mean?

INTERPRETATION

Confidence level is a repeated-sampling property



Before sampling

The interval is random because it depends on Y .

After sampling

The realized interval either contains the fixed parameter or it does not.

Coverage probability belongs to the procedure.

A common misconception

PITFALL

Incorrect

"There is a 95% probability that β is in this interval."

Frequentist statement

"This procedure produces intervals that contain β with probability 0.95 over repeated samples."

Once the data are observed, β is fixed and the endpoints are fixed.

Confidence intervals are inverted hypothesis tests

DUALITY

For every candidate θ_0 , test $H_0 : c^T \beta = \theta_0$

A two-sided level- α t test rejects when

Solve the non-rejection inequality for θ_0 :

The set of null values not rejected is exactly the $(1 - \alpha)$ confidence interval.

Rank-deficient models: interval targets must be estimable

ESTIMABILITY

If $\text{rank}(X) < p$, β itself is not uniquely identified.

$$c^T \beta \text{ is estimable} \iff c \in \mathcal{C}(X^T)$$

Estimable target

Its value is invariant across all β that produce the same mean $X\beta$. A valid confidence interval can be constructed for it.

Non-estimable target

Different parameter vectors compatible with exactly the same distribution of Y imply different target values. No data-based interval can resolve that ambiguity.

The inference should be expressed in terms of identifiable functions, not an arbitrary identifying constraint.

02

Confidence regions

One interval protects one direction. A region protects several parameter directions jointly.

Known σ : confidence region for β

ELLIPSOID

Standardize the estimation error using its covariance metric:

$$\frac{(\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta)}{\sigma^2} \sim \chi_p^2$$

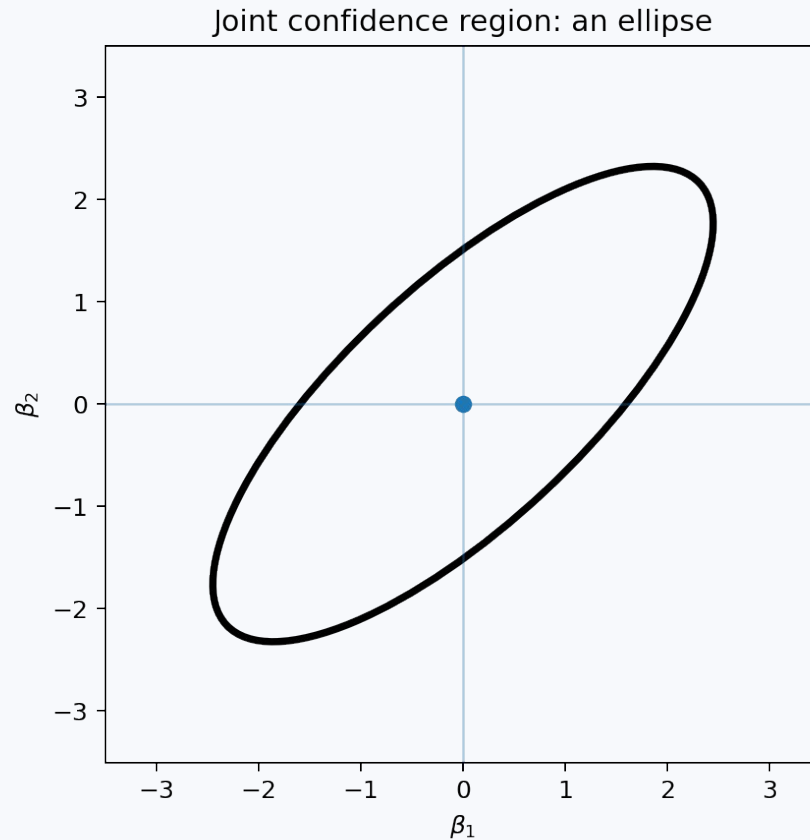
Therefore a $(1 - \alpha)$ confidence region is

$$\mathcal{C}_{1-\alpha} = \left\{ \beta : (\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta) \leq \sigma^2 \chi_{p,1-\alpha}^2 \right\}$$

The shape is an ellipsoid centered at $\hat{\beta}$.

Geometry of a confidence region

GEOMETRY



Center $\hat{\beta}$

Orientation

Eigenvectors of $X^T X$ (equivalently of the covariance).

Axis lengths Longer in weakly informed directions.

Unknown σ : the exact F confidence region

F REGION

Because $\hat{\beta}$ and s^2 are independent,

$$\frac{(\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta) / p}{s^2} \sim F_{p, n-p}$$

$$(\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta) \leq p s^2 F_{p, n-p, 1-\alpha}$$

This is the multivariate analogue of replacing z by t in a scalar interval.

Why does the joint region use F rather than t?

INTUITION

One direction

A scalar contrast projects the estimation error onto one line. After standardization, the numerator is one squared normal direction, so the signed pivot is t.

p directions jointly

The confidence region measures squared distance across p independent parameter directions. The numerator is a p-df chi-squared quantity; dividing by s^2 produces F.

$$t_{\nu}^2 = F_{1,\nu}$$

The scalar case is contained inside the general region theory.

The confidence region is the inversion of a global F test

DUALITY

For a candidate β_0 , test $H_0 : \beta = \beta_0$

The corresponding F statistic is
$$F(\beta_0) = \frac{(\hat{\beta} - \beta_0)^T X^T X (\hat{\beta} - \beta_0) / p}{s^2}$$

Confidence region = all β_0 for which $F(\beta_0) \leq F_{p, n-p, 1-\alpha}$

Testing and confidence regions are the same geometry viewed from opposite directions.

Confidence region for several linear functions $C\beta$

GENERAL HYPOTHESIS

Let C be $q \times p$ with rank q . Define $\theta = C\beta$.

$$C\hat{\beta} \sim N_q(C\beta, \sigma^2 C(X^T X)^{-1} C^T)$$

With σ unknown, the exact $(1 - \alpha)$ region is

$$(C\hat{\beta} - C\beta)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - C\beta) \leq q s^2 F_{q, n-p, 1-\alpha}$$

Only the q -dimensional target matters; nuisance parameter directions disappear.

When $q = 1$, the region collapses to the usual t interval

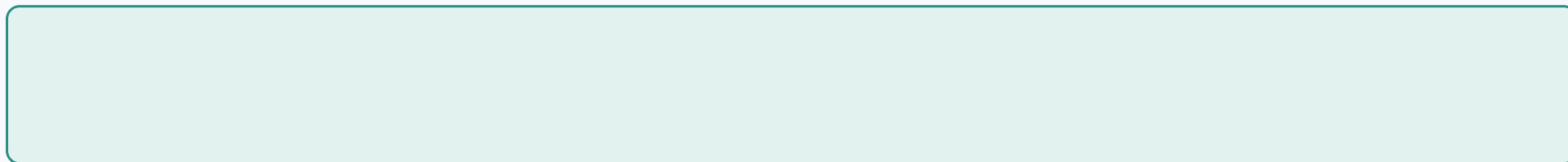
CONSISTENCY CHECK

$$\frac{(c^T \hat{\beta} - c^T \beta)^2}{s^2 c^T (X^T X)^{-1} c} \leq F_{1, n-p, 1-\alpha}$$

But

$$F_{1, \nu, 1-\alpha} = t_{\nu, 1-\alpha/2}^2$$

Taking square roots gives



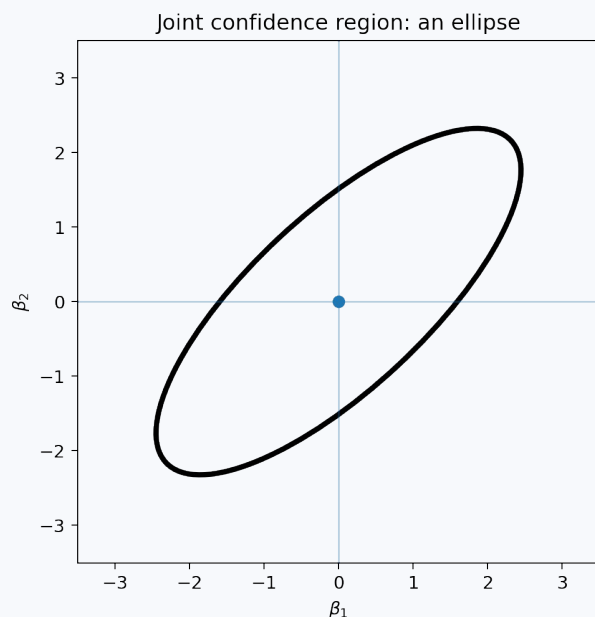
The scalar and multivariate theories match exactly.

Worked example: a two-parameter joint region

EXAMPLE

Suppose $q = 2$, $n - p = 30$, $s^2 = 4$, and $F_{2,30,0.95} \approx 3.32$.

$$(\hat{\theta} - \theta)^T V^{-1} (\hat{\theta} - \theta) \leq 2(4)(3.32) = 26.56$$



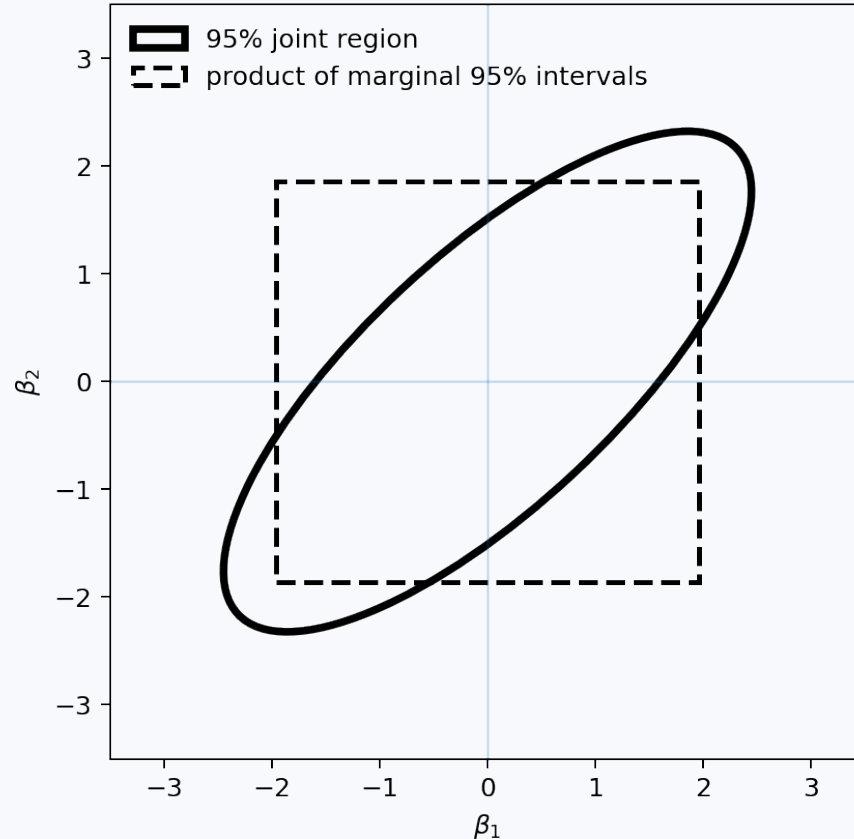
Interpretation

The joint region controls coverage of the 2-vector θ as a whole. Correlation between its components determines the ellipse orientation.

Marginal intervals do not automatically give joint 95% coverage

JOINT COVERAGE

Marginal intervals are not a 95% joint statement



Two 95% marginals

Each protects one coordinate separately. The probability that both cover is generally < 0.95 .

A 95% joint region

Protects the entire parameter vector simultaneously with probability 0.95.

Joint statements require joint calibration.

03

Simultaneous inference

When we plan to inspect many contrasts, pointwise 95% intervals are no longer enough.

Pointwise, joint, and simultaneous are different goals

COVERAGE

Pointwise

One prespecified contrast $c^T\beta$.
Coverage: $1 - \alpha$ for that single target.

Joint

A finite-dimensional vector $C\beta$.
Coverage: $1 - \alpha$ for the whole vector.

Simultaneous

A family of contrasts, possibly all c .
Coverage: $1 - \alpha$ for every member at once.

Choose the inferential target first; then choose the calibration.

Bonferroni simultaneous intervals

BONFERRONI

Suppose we have m prespecified estimable contrasts $c_1^T\beta, \dots, c_m^T\beta$.

Use individual error level α/m , giving

$$c_j^T \hat{\beta} \pm t_{n-p, 1-\alpha/(2m)} s \sqrt{c_j^T (X^T X)^{-1} c_j}$$

Strength

Simple, transparent, and useful when the family of comparisons is small and prespecified.

Cost

Can become conservative as m grows because it protects each interval using a stricter cutoff.

Scheffé simultaneous intervals protect all linear combinations

SCHEFFE

The joint F region implies a simultaneous bound valid for every $c \in \mathbb{R}^p$:

$$c^T \hat{\beta} \pm \sqrt{p F_{p, n-p, 1-\alpha}} s \sqrt{c^T (X^T X)^{-1} c}$$

simultaneously for all c

Why it works

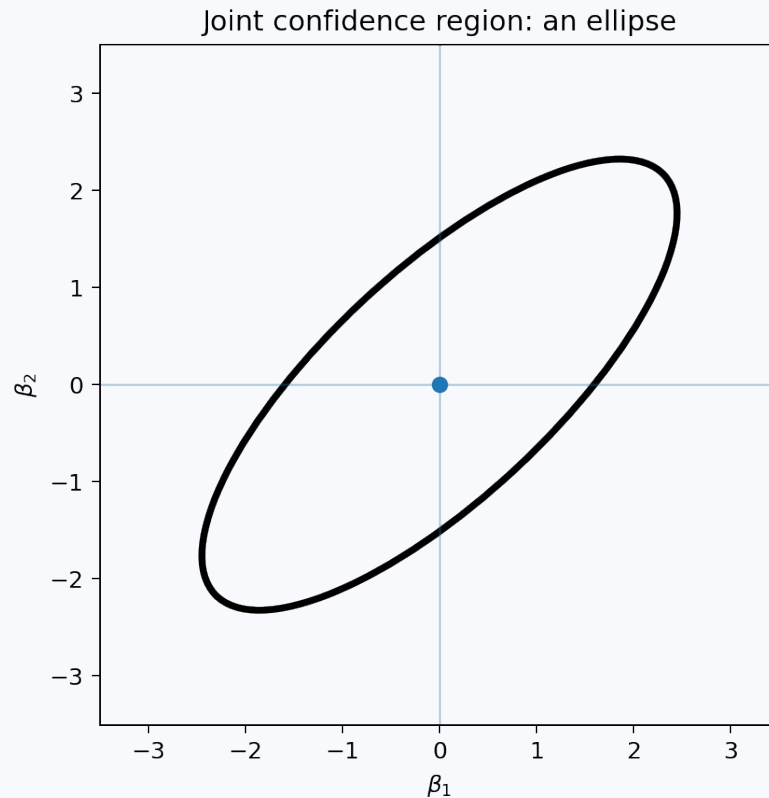
If β lies inside the joint ellipsoid, then every linear projection $c^T \beta$ lies inside its induced interval.

When useful

Exploratory post hoc comparisons, all contrasts among means, or any situation where the direction c is chosen after seeing the data.

Scheffé intervals are projections of the confidence ellipsoid

GEOMETRY



Pick any direction c .

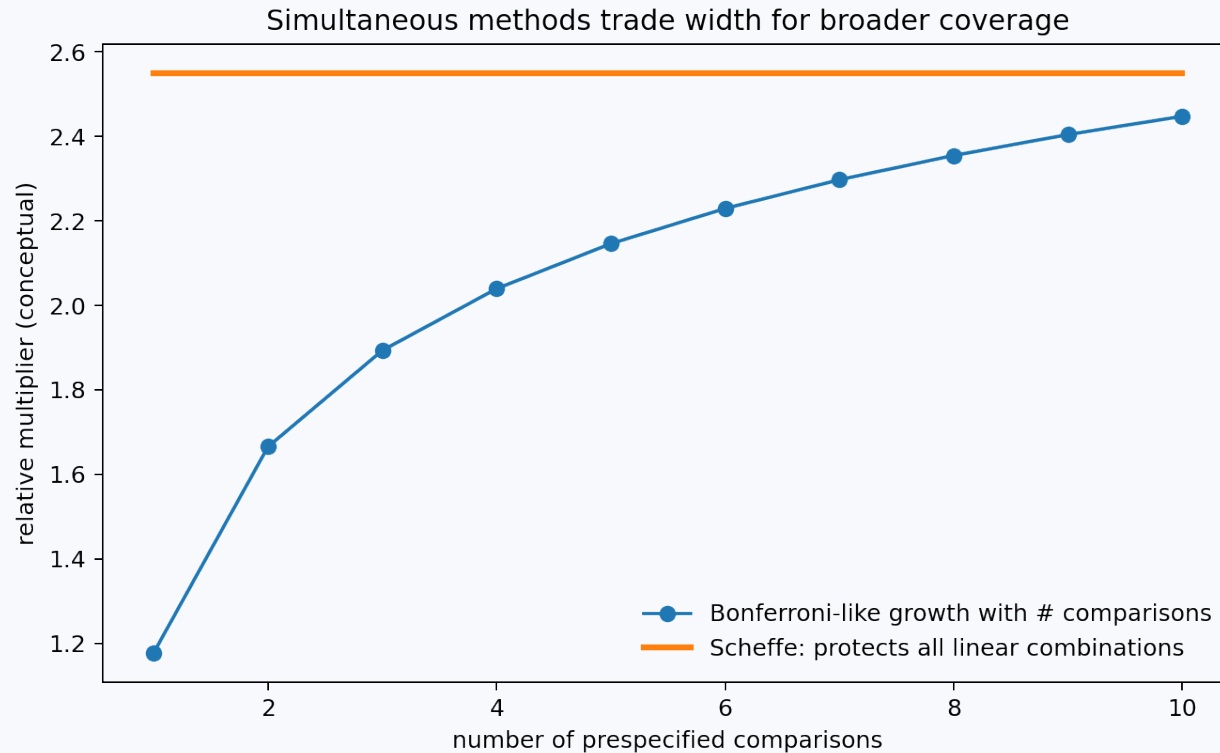
Project the entire ellipsoid onto that direction.

The extreme projected values become the simultaneous confidence limits for $c^T\beta$.

One ellipsoid $\rightarrow$ infinitely many protected linear contrasts

Bonferroni vs Scheffé

COMPARISON



Bonferroni

Best suited to a finite, prespecified collection of comparisons.

Scheffé

Protects every linear combination, including post hoc choices; often wider for a small family.

There is no universally best method—the family you want to protect determines the calibration.

Mean response and prediction

The same linear-model machinery gives uncertainty for fitted means and future observations.

Confidence interval for the mean response at x_0

MEAN RESPONSE

The mean response is an estimable linear function:

$$\mu_0 = E(Y \mid x_0) = x_0^T \beta, \quad \hat{\mu}_0 = x_0^T \hat{\beta}$$

$$SE(\hat{\mu}_0) = s \sqrt{x_0^T (X^T X)^{-1} x_0}$$

$$x_0^T \hat{\beta} \pm t_{n-p, 1-\alpha/2} s \sqrt{x_0^T (X^T X)^{-1} x_0}$$

This interval targets the conditional mean—not an individual future outcome.

Prediction interval for a new observation

PREDICTION

For a future independent observation $Y_0 = x_0^T \beta + \varepsilon_0$, $\varepsilon_0 \sim N(0, \sigma^2)$

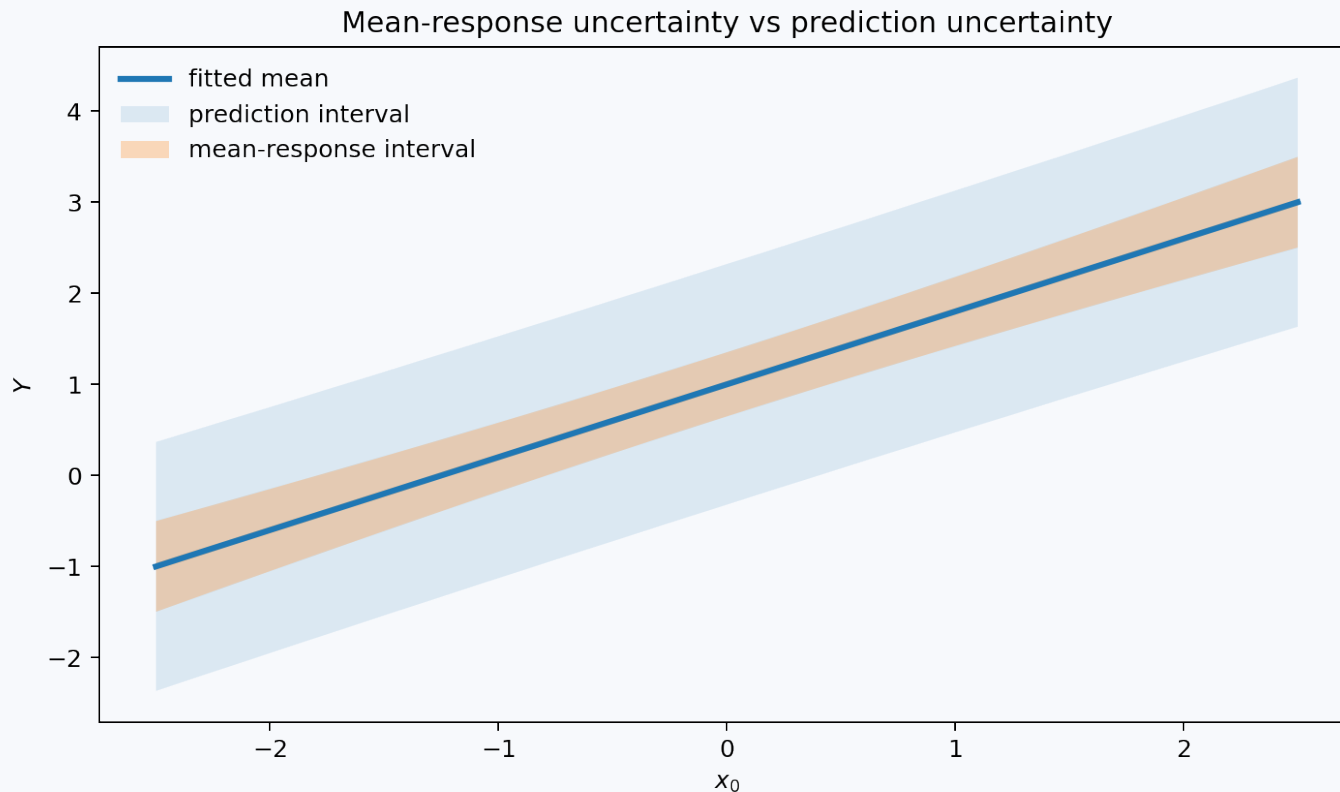
Prediction error: $Y_0 - x_0^T \hat{\beta} = \varepsilon_0 - x_0^T (\hat{\beta} - \beta)$

Variance: $\sigma^2 [1 + x_0^T (X^T X)^{-1} x_0]$

$$x_0^T \hat{\beta} \pm t_{n-p, 1-\alpha/2} s \sqrt{1 + x_0^T (X^T X)^{-1} x_0}$$

Why prediction intervals are wider

VARIANCE DECOMPOSITION



Mean-response uncertainty

Only uncertainty in estimating $x_0^T \beta$.

Prediction uncertainty

Parameter uncertainty + new irreducible noise ε_0 . The extra "1" is the future observation's own variance.

The interval width is controlled by leverage at x_0

DESIGN GEOMETRY

$$h(x_0) = x_0^T (X^T X)^{-1} x_0$$

Mean-response CI

Half-width $\propto \sqrt{h(x_0)}$.

Small near well-supported parts of the design; large in poorly supported directions.

Prediction interval

Half-width $\propto \sqrt{1 + h(x_0)}$.

Even with perfect knowledge of the mean, irreducible outcome noise remains.

Extrapolation increases $h(x_0)$, so both intervals widen.

A simultaneous confidence band for the regression mean

WORKING-HOTELLING

In simple linear regression, β has dimension $p = 2$. Scheffé gives a band that protects the entire fitted line simultaneously:

$$\hat{y}(x) \pm \sqrt{2F_{2,n-2,1-\alpha}} s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}$$

Pointwise band

At each fixed x , use t_{n-2} .

Simultaneous band

Protects the whole line for all x at once.

The simultaneous band is wider because it answers a stronger question.

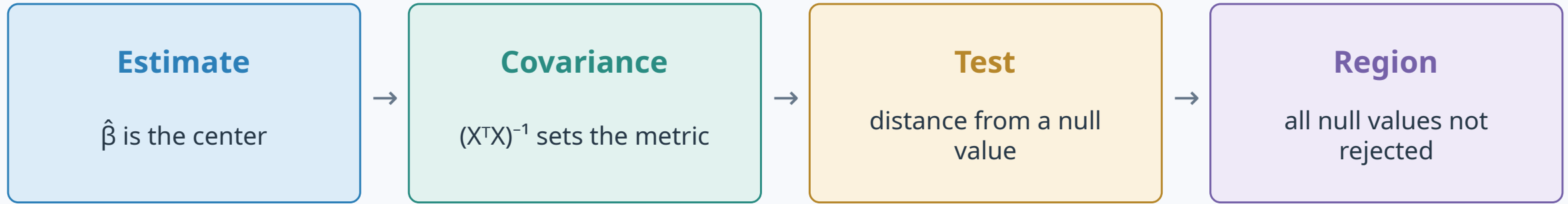
05

Connections and summary

Confidence procedures are another face of the same projection geometry used for tests and sums of squares.

One geometric object links estimation, testing, and confidence regions

UNIFYING VIEW



quadratic distance in parameter space $\iff$ F statistic

The confidence ellipsoid is simply a contour of constant F statistic around $\hat{\beta}$.

A compact confidence-procedure table

SUMMARY			
Target	Standard error / metric	Calibration	Procedure
$c^T\beta$	$s\sqrt{c^T(X^TX)^{-1}c}$	t_{n-p}	Pointwise CI
β	X^TX metric	$F_{p,n-p}$	Joint ellipsoid
$C\beta$	$[C(X^TX)^{-1}C^T]^{-1}$ metric	$F_{q,n-p}$	q-dim joint region
m fixed contrasts	contrast-specific SE	Bonferroni t	Simultaneous CIs
all $c^T\beta$	contrast-specific SE	$\sqrt{pF_{p,n-p}}$	Scheffé CIs
mean at x_0	$s\sqrt{x_0^T(X^TX)^{-1}x_0}$	t_{n-p}	Mean-response CI
future Y_0	$s\sqrt{1+x_0^T(X^TX)^{-1}x_0}$	t_{n-p}	Prediction interval

Common mistakes to avoid

PITFALLS

1. Confusing marginal and joint coverage

Several 95% intervals do not automatically form a 95% confidence statement for the whole vector.

2. Interpreting 95% as posterior probability

In frequentist inference the parameter is fixed; the procedure has 95% repeated-sampling coverage.

3. Ignoring estimability

In a rank-deficient model, inference must target identifiable linear functions.

4. Using a mean-response CI to predict individuals

Prediction adds new-observation noise and is necessarily wider.

Practice: choose the correct confidence procedure

PRACTICE

- You want a 95% interval for β_3 only. Which pivot and degrees of freedom are used?
- You want a 95% joint statement about (β_2, β_3) . What distribution calibrates the region?
- You plan to inspect every contrast among six treatment means after fitting the model. Why might Scheffé be preferable to ordinary t intervals?
- At x_0 , what is the difference between a confidence interval for $E(Y | x_0)$ and a prediction interval for Y_0 ?
- In a rank-deficient model, what condition must c satisfy before a confidence interval for $c^T\beta$ is meaningful?

Takeaways

SUMMARY

Scalar CI

A t pivot gives exact finite-sample intervals for estimable linear functions.

Joint region

Quadratic distance in the covariance metric produces an F-calibrated ellipsoid.

Duality

A confidence set is the collection of parameter values not rejected by the corresponding test.

Simultaneous inference

Bonferroni protects a finite family; Scheffé protects all linear combinations.

Prediction

Future observations add irreducible noise beyond uncertainty in the fitted mean.

one normal model + projection geometry + pivots $\Rightarrow$ a unified theory of uncertainty